Assignment-1
Embedded systems and IoT Application
BET-C620

Submission Last Date: 27/02/2026

Note: Attempt any 5 question from Section A, any 3 question from section B and any 3 from section C.

Section A (Short Answer Questions)

1. Define an Embedded System. How is it different from a general-purpose computer?
2. List and explain any four applications of embedded systems in daily life.
3. What is an Embedded Operating System? Mention its key features.
4. Explain the concept of real-time operating systems (RTOS) in embedded systems.
5. What are the major design parameters of an embedded system?
6. Define design life cycle in embedded system development.
7. What is hardware and software partitioning? Why is it important?
8. Define co-design in embedded systems.
9. What is an open collector output? State its applications.
10. What is tristate output? Why is it required in digital systems?

Section B (Descriptive Questions)

11. Explain in detail the architecture of an embedded system with a neat block diagram.
12. Discuss the complete design life cycle of an embedded system.
13. Explain hardware–software co-design methodology with suitable examples.
14. Discuss various design parameters of an embedded system and their significance in real-world applications.
15. Explain the concept of I/O sink and I/O sourcing with suitable diagrams.

Section C (Long Answer / Analytical Questions)

16. Explain Programmable Logic Devices (PLDs). Compare different types of PLDs.
17. Describe the working principle and importance of a Watchdog Timer in embedded systems.
18. Discuss hardware design and development process in embedded systems.
19. Explain the role of embedded operating systems in improving system performance and reliability.
20. Design an embedded system for a smart home application and explain hardware and software partitioning involved.